

Numerical Analysis homework # 5

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Finish time: December 7, 2025

I. Question I

Verification of Vector Space Axioms (Definition B.2)

Let $u, v, w \in \mathcal{C}[a, b]$ and $a, b \in \mathbb{C}$. Since the sum of continuous functions is continuous and the scalar multiple of a continuous function is continuous, $\mathcal{C}[a, b]$ is closed under these operations. We verify the axioms pointwise for all $t \in [a, b]$:

1. **(VSA-1) Commutativity:** $u(t) + v(t) = v(t) + u(t)$ holds by the commutativity of addition in $\mathbb{C}$. Thus, $u + v = v + u$.
2. **(VSA-2) Associativity:** $(u(t) + v(t)) + w(t) = u(t) + (v(t) + w(t))$ holds by the associativity of addition in $\mathbb{C}$. Thus, $(u + v) + w = u + (v + w)$.
3. **(VSA-3) Compatibility:** $(ab)u(t) = a(bu(t))$ holds by the associativity of multiplication in $\mathbb{C}$. Thus, $(ab)u = a(bu)$.
4. **(VSA-4) Additive Identity:** Let 0 be the zero function. Since constant functions are continuous, $0 \in \mathcal{C}[a, b]$. Clearly, $u(t) + 0 = u(t)$. Thus, $u + 0 = u$.
5. **(VSA-5) Additive Inverse:** For any $u \in \mathcal{C}[a, b]$, let $-u$ be defined by $(-u)(t) = -u(t)$. This is continuous, and $u(t) + (-u(t)) = 0$. Thus, $u + (-u) = 0$.
6. **(VSA-6) Multiplicative Identity:** Let $1 \in \mathbb{C}$. Then $1 \cdot u(t) = u(t)$. Thus, $1u = u$.
7. **(VSA-7) Distributive Laws:** By the distributivity of complex numbers: $(a + b)u(t) = au(t) + bu(t)$ implies $(a + b)u = au + bu$. $a(u(t) + v(t)) = au(t) + av(t)$ implies $a(u + v) = au + av$.

Verification of Inner Product Axioms (Definition B.165)

We verify the properties for $\langle u, v \rangle = \int_a^b \rho(t)u(t)\overline{v(t)} dt$, given $\rho(t) > 0$.

1. **(IP-1) Real Positivity:** $\langle u, u \rangle = \int_a^b \rho(t)|u(t)|^2 dt$. Since $\rho(t) > 0$ and $|u(t)|^2 \geq 0$, the integral is non-negative. $\langle u, u \rangle \geq 0$.
2. **(IP-2) Definiteness:** $\langle u, u \rangle = 0 \iff \int_a^b \rho(t)|u(t)|^2 dt = 0$. Since $\rho(t)|u(t)|^2$ is non-negative and continuous, the integral is zero if and only if the function is identically zero. As $\rho(t) > 0$, this implies $|u(t)|^2 = 0$, from which we can obtain that $u(t) \equiv 0$.
3. **(IP-3) Additivity in the first slot:** $\langle u + v, w \rangle = \int \rho(u + v)\overline{w} = \int \rho u\overline{w} + \int \rho v\overline{w} = \langle u, w \rangle + \langle v, w \rangle$ by the linearity of integration.
4. **(IP-4) Homogeneity in the first slot:** $\langle au, w \rangle = \int \rho(au)\overline{w} = a \int \rho u\overline{w} = a\langle u, w \rangle$.
5. **(IP-5) Conjugate Symmetry:** $\overline{\langle v, u \rangle} = \overline{\int \rho v\overline{u}} = \int \overline{\rho v\overline{u}} = \int \rho \overline{v}u = \int \rho u\overline{v} = \langle u, v \rangle$ (using $\rho \in \mathbb{R}$).

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Verification of Normed Vector Space (Definition B.170)

We define the induced norm $\|u\|_2 = \sqrt{\langle u, u \rangle}$ (Definition B.170) and verify it satisfies the standard norm axioms:

1. **Positivity and Definiteness:** From (IP-1) and (IP-2), $\|u\|_2 \geq 0$ and $\|u\|_2 = 0 \iff u = 0$.
2. **Homogeneity:** $\|au\|_2^2 = \langle au, au \rangle = a\bar{a}\langle u, u \rangle = |a|^2\|u\|_2^2$. from which we can obtain $\|au\|_2 = |a|\|u\|_2$.
3. **Triangle Inequality:** We must show $\|u + v\|_2 \leq \|u\|_2 + \|v\|_2$. Expansion of the squared norm gives:

$$\|u + v\|_2^2 = \langle u + v, u + v \rangle = \|u\|_2^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} + \|v\|_2^2 = \|u\|_2^2 + 2\text{Re}(\langle u, v \rangle) + \|v\|_2^2.$$

By Cauchy-Schwarz inequality, $|\langle u, v \rangle| \leq \|u\|_2\|v\|_2$. Since $\text{Re}(z) \leq |z|$, we have:

$$\|u + v\|_2^2 \leq \|u\|_2^2 + 2\|u\|_2\|v\|_2 + \|v\|_2^2 = (\|u\|_2 + \|v\|_2)^2.$$

Taking the square root gives $\|u + v\|_2 \leq \|u\|_2 + \|v\|_2$.

Since $\mathcal{C}[a, b]$ satisfies all axioms for a vector space, an inner product space, and the induced norm satisfies the norm axioms, Theorem 5.7 is proved.

II. Question II

From Definition 2.42, the Chebyshev polynomial of degree $n \in \mathbb{N}$ of the first kind is a polynomial $T_n : [-1, 1] \rightarrow [-1, 1]$ and we have

$$T_n(x) = \cos(n \arccos x) \quad (1)$$

II.a

Through Theorem 5.7, we have

$$\langle T_i, T_j \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \cos(i \arccos x) \cos(j \arccos x) dx \quad (2)$$

in which $\cos(i \arccos x) \cos(j \arccos x) = \frac{1}{2}[\cos((i+j) \arccos x) + \cos((i-j) \arccos x)]$. Thus, we have

- when $i = j = 0$, we can obtain that

$$\langle T_i, T_j \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \arcsin x \Big|_{-1}^1 = \pi \quad (3)$$

- when $i = j \neq 0$, we have

$$\langle T_i, T_j \rangle = \int_{-1}^1 \frac{1}{2\sqrt{1-x^2}} [\cos((i+j) \arccos x) + 1] dx = \int_{-1}^1 \frac{1}{2\sqrt{1-x^2}} dx = \frac{\pi}{2} \quad (4)$$

- when $i \neq j$, we have

$$\langle T_i, T_j \rangle = \int_{-1}^1 \frac{1}{2\sqrt{1-x^2}} [\cos((i+j) \arccos x) + \cos((i-j) \arccos x)] dx = 0 \quad (5)$$

So $\langle T_i, T_j \rangle > 0$ holds when $i = j$, while $\langle T_i, T_j \rangle = 0$ holds when $i \neq j$, which means that they are orthogonal on $[-1, 1]$.

II.b

From II.a we have known that T_i and T_j is orthogonal, so what we need to do is to normalize them. Since $\|T_n\|_2 = \sqrt{\langle T_n, T_n \rangle}$, we can obtain that:

$$\|T_n\|_2 = \begin{cases} \sqrt{\frac{\pi}{2}}, & n \neq 0 \\ \sqrt{\pi}, & n = 0 \end{cases} \quad (6)$$

So it is easy to obtain that

$$T_0^* = \frac{T_0}{\|T_0\|_2} = \frac{1}{\sqrt{\pi}} \quad (7)$$

$$T_1^* = \frac{T_1}{\|T_1\|_2} = \sqrt{\frac{2}{\pi}}x \quad (8)$$

$$T_2^* = \frac{T_2}{\|T_2\|_2} = \sqrt{\frac{2}{\pi}}(2x^2 - 1) \quad (9)$$

Then $\{T_0^*, T_1^*, T_2^*\}$ arrive at an orthonormal system.

III. Question III

III.a

From Question II we have obtained a set of Chebyshev polynomial orthonormal systems T_0^*, T_1^*, T_2^* , in which T_0^*, T_1^*, T_2^* satisfies equation (7)(8)(9). And with Fourier expansion, we have

$$y(x) = \sum_{i=0}^2 (y, T_i^*) T_i^* \quad (10)$$

so we compute (y, T_i^*) as follows:

$$(y, T_0^*) = \int_{-1}^1 \frac{1}{\sqrt{\pi}} dx = \frac{2}{\sqrt{\pi}} \quad (11)$$

$$(y, T_1^*) = \int_{-1}^1 \sqrt{\frac{2}{\pi}} x dx = 0 \quad (12)$$

$$(y, T_2^*) = \int_{-1}^1 \sqrt{\frac{2}{\pi}} (2x^2 - 1) dx = -\frac{2}{3} \sqrt{\frac{2}{\pi}} \quad (13)$$

Thus we have

$$y(x) = \frac{2}{\pi} - \frac{2}{3} \sqrt{\frac{2}{\pi}} \times \sqrt{\frac{2}{\pi}} (2x^2 - 1) = -\frac{8}{3\pi} x^2 + \frac{10}{3\pi} \quad (14)$$

III.b

By using the single polynomial independent list $\{1, x, x^2\}$, first, we compute the Gram matrix $G(1, x, x^2)$. In the integral as follows:

$$\int_{-1}^1 \frac{x^k}{\sqrt{1-x^2}} dx \quad (15)$$

we denote $x = \cos \theta$, then $dx = -\sin \theta d\theta$, then we can obtain that

$$\int_{-1}^1 \frac{x^k}{\sqrt{1-x^2}} dx = \int_{\pi}^0 \frac{\cos^k \theta}{\sin \theta} (-\sin \theta) d\theta = \int_0^{\pi} \cos^k \theta d\theta \quad (16)$$

From which we can obtain that

$$\langle 1, 1 \rangle = \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = \pi \quad (17)$$

$$\langle 1, x \rangle = \int_{-1}^1 \frac{x}{\sqrt{1-x^2}} dx = 0 \quad (18)$$

$$\langle 1, x^2 \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2} \quad (19)$$

$$\langle x, x \rangle = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{\pi}{2} \quad (20)$$

$$\langle x, x^2 \rangle = \int_{-1}^1 \frac{x^3}{\sqrt{1-x^2}} dx = 0 \quad (21)$$

$$\langle x^2, x^2 \rangle = \int_{-1}^1 \frac{x^4}{\sqrt{1-x^2}} dx = \frac{3\pi}{8} \quad (22)$$

Then the Gram matrix will be

$$G(1, x, x^2) = \begin{bmatrix} \pi & 0 & \frac{\pi}{2} \\ 0 & \frac{\pi}{2} & 0 \\ \frac{\pi}{2} & 0 & \frac{3\pi}{8} \end{bmatrix} \quad (23)$$

while $c = [\langle y, 1 \rangle, \langle y, x \rangle, \langle y, x^2 \rangle]^T = [2, 0, \frac{2}{3}]^T$, then we solve the equation $G^T a = c$, from which we can obtain that

$$a = [\frac{10}{3\pi}, 0, \frac{-8}{3\pi}] \quad (24)$$

which means that $y(x) = -\frac{8}{3\pi}x^2 + \frac{10}{3\pi}$

IV. Question IV

IV.a

We are given $N = 12$ data points with $x_i \in \{1, 2, \dots, 12\}$ and the weight function $\rho(x) = 1$. The inner product is defined as:

$$\langle u, v \rangle = \sum_{i=1}^{12} u(x_i) v(x_i) \quad (25)$$

We apply the Gram-Schmidt process to the basis $\{1, x, x^2\}$. First we compute the norm squared:

$$\|u_0\|^2 = \langle 1, 1 \rangle = \sum_{i=1}^{12} 1 = 12$$

Thus, the first orthonormal polynomial is:

$$u_0^*(t) = \frac{1}{\sqrt{12}} \quad (26)$$

Then we compute the projection of x onto u_0^* :

$$\langle x, u_0^* \rangle = \sum_{i=1}^{12} x_i \cdot \frac{1}{\sqrt{12}} = \frac{1}{\sqrt{12}} \cdot \frac{12(13)}{2} = \frac{78}{\sqrt{12}}$$

and then construct the orthogonal vector v_1 :

$$v_1(t) = x - \langle x, u_0^* \rangle u_0^*(t) = x - \frac{78}{\sqrt{12}} \cdot \frac{1}{\sqrt{12}} = x - \frac{78}{12} = x - 6.5$$

then the norm squared of v_1 :

$$\|v_1\|^2 = \sum_{i=1}^{12} (x_i - 6.5)^2$$

Since the points are symmetric around 6.5, this sum is:

$$2 \sum_{k=0}^5 (0.5 + k)^2 = 2(0.5^2 + 1.5^2 + 2.5^2 + 3.5^2 + 4.5^2 + 5.5^2) = 2(71.5) = 143$$

Thus, the second orthonormal polynomial is:

$$u_1^*(t) = \frac{x - 6.5}{\sqrt{143}} \quad (27)$$

We compute projections onto previous vectors. Using $\sum x_i^2 = 650$ and $\sum x_i^3 = 6084$:

$$\begin{aligned} \langle x^2, u_0^* \rangle &= \frac{1}{\sqrt{12}} \sum x_i^2 = \frac{650}{\sqrt{12}} \\ \langle x^2, u_1^* \rangle &= \frac{1}{\sqrt{143}} \sum x_i^2 (x_i - 6.5) = \frac{1}{\sqrt{143}} (6084 - 6.5 \cdot 650) = \frac{1859}{\sqrt{143}} \end{aligned}$$

Note that $\frac{1859}{143} = 13$, so the coefficient simplifies nicely in later steps. thus we construct v_2 :

$$\begin{aligned} v_2(t) &= x^2 - \frac{650}{\sqrt{12}}u_0^*(t) - \frac{1859}{\sqrt{143}}u_1^*(t) \\ &= x^2 - \frac{650}{12} - 13(x - 6.5) \\ &= x^2 - 13x + (84.5 - 54.166\dots) \\ &= x^2 - 13x + \frac{91}{3} \end{aligned}$$

then we compute the norm squared $\|v_2\|^2$. A direct summation or the property $\|v_2\|^2 = \|x^2\|^2 - |\langle x^2, u_0^* \rangle|^2 - |\langle x^2, u_1^* \rangle|^2$ using $\sum x^4 = 60710$ yields:

$$\|v_2\|^2 = \frac{4004}{3}$$

Thus, the third orthonormal polynomial is:

$$u_2^*(t) = \frac{x^2 - 13x + \frac{91}{3}}{\sqrt{\frac{4004}{3}}} = \sqrt{\frac{3}{4004}}(x^2 - 13x + \frac{91}{3}) \quad (28)$$

So the orthonormal polynomials are $\{\frac{1}{2\sqrt{3}}, \frac{x-\frac{13}{2}}{\sqrt{143}}, \sqrt{\frac{3}{4004}}(x^2 - 13x + \frac{91}{3})\}$

IV.b

We calculate the Fourier coefficients $c_k = \langle y, u_k^* \rangle$ using the data sums: $\sum y_i = 1662$, $\sum x_i y_i = 11392$, and $\sum x_i^2 y_i = 109750$, we can obtain that

- $c_0 = \langle y, u_0^* \rangle = \frac{1}{\sqrt{12}} \sum y_i = \frac{1662}{\sqrt{12}}$
- $c_1 = \langle y, u_1^* \rangle = \frac{1}{\sqrt{143}} (\sum x_i y_i - 6.5 \sum y_i) = \frac{11392 - 10803}{\sqrt{143}} = \frac{589}{\sqrt{143}}$
- $c_2 = \langle y, u_2^* \rangle = \frac{1}{\|v_2\|} (\sum x_i^2 y_i - 13 \sum x_i y_i + \frac{91}{3} \sum y_i)$

And by calculating

$$109750 - 13(11392) + \frac{91}{3}(1662) = 109750 - 148096 + 50414 = 12068.$$

we have $c_2 = \sqrt{\frac{3}{4004}} 12068$. While the best approximation is $\hat{\varphi}(t) = c_0 u_0^*(t) + c_1 u_1^*(t) + c_2 u_2^*(t)$. We rewrite this in the form $a_0 + a_1 x + a_2 x^2$.

Then let's consider the coefficient of x^2 , which we note a_2 , it is easy to know that only u_2^* contains x^2 .

$$a_2 = \frac{c_2}{\|v_2\|} = \frac{12068}{4004/3} = \frac{36204}{4004} \approx 9.0419.$$

Then coefficient of x , which we called a_1 , contributions are from u_1^* and u_2^* , so we have

$$a_1 = \frac{c_1}{\sqrt{143}} - 13a_2 = \frac{589}{143} - 13(9.0419) = 4.1188 - 117.545 \approx -113.43.$$

Then constant term a_0 , Contributions are from all three, we have

$$a_0 = \frac{c_0}{\sqrt{12}} - 6.5 \frac{c_1}{\sqrt{143}} + \frac{91}{3} a_2 \approx 386.00$$

So the computed polynomial is:

$$\hat{\varphi}(x) = 386.00 - 113.43x + 9.04x^2.$$

These values exactly match the result from the Normal Equations method in Example 5.55 ($a = [386.00, -113.43, 9.04]^\top$), verifying the correctness of the orthonormal polynomial approach.

IV.c

1. Reusable Calculations The construction of the orthonormal system actually depends only on the inner product space definition, which is determined by:

- The sampling points x_i (the domain),
- The weight function $\rho(x)$ (here $\rho = 1$),
- The original linearly independent basis (here $\{1, x, x^2\}$).

Thus it does not depend on the target values y_i . Therefore, the orthonormal polynomials $u_0^*(x), u_1^*(x), u_2^*(x)$ can be fully reused for any new dataset y .

2. Non-Reusable Calculations Since the target vector y changes, any quantity linearly dependent on y must be recomputed:

- The Fourier coefficients $c_k = \langle y, u_k^* \rangle = \sum_{i=1}^N y_i u_k^*(x_i)$.
- The final best approximation coefficients a_i in the standard basis form $\sum a_i x^i$.

In my opinion, the reuse of orthonormal polynomials highlights a critical advantage regarding Numerical Stability and Conditioning:

- Ill-conditioning of Normal Equations: In the Normal Equation approach ($Ga = c$), the Gram matrix G is formed by the monomial basis $\{1, x, x^2, \dots\}$. As noted in Example 5.42, this matrix resembles the Hilbert matrix, which is notoriously ill-conditioned. Even if G (or its inverse) is pre-computed and reused, the operation involves a matrix with a very high condition number, making the solution highly sensitive to perturbations or rounding errors in y .
- Perfect Conditioning of Orthonormal Systems: By reusing the pre-computed orthonormal basis $\{u_k^*\}$, the corresponding Gram matrix is the Identity matrix I . The condition number is exactly 1. Finding the new best approximation reduces to simple orthogonal projections, which are numerically stable operations.
- The calculation of each coefficient c_k is independent of the others. If we were to increase the degree of the polynomial for a new fit, previous coefficients would remain valid. In the Normal Equations, changing the degree requires solving a completely new linear system.

V. Question V

Let (σ_k, u_k, v_k) be the singular system of A with rank r . We can express A and its pseudo-inverse A^+ as

$$A = \sum_{k=1}^r \sigma_k v_k u_k^* \quad \text{and} \quad A^+ = \sum_{k=1}^r \frac{1}{\sigma_k} u_k v_k^* \quad (29)$$

Proof of Theorem 5.66

First, we verify the Hermitian properties. We have

$$AA^+ = \left(\sum_{j=1}^r \sigma_j v_j u_j^* \right) \left(\sum_{k=1}^r \frac{1}{\sigma_k} u_k v_k^* \right) = \sum_{j=1}^r \sum_{k=1}^r \frac{\sigma_j}{\sigma_k} v_j (u_j^* u_k) v_k^* \quad (30)$$

Using the orthogonality condition $u_j^* u_k = \delta_{jk}$, we obtain

$$AA^+ = \sum_{k=1}^r v_k v_k^* \quad (31)$$

Since $(v_k v_k^*)^* = v_k v_k^*$, AA^+ is Hermitian. Similarly, using $v_j^* v_k = \delta_{jk}$, we have

$$A^+ A = \left(\sum_{j=1}^r \frac{1}{\sigma_j} u_j v_j^* \right) \left(\sum_{k=1}^r \sigma_k v_k u_k^* \right) = \sum_{k=1}^r u_k u_k^* \quad (32)$$

Since $(u_k u_k^*)^* = u_k u_k^*$, $A^+ A$ is Hermitian. Next, we verify that $AA^+ A = A$ and $A^+ AA^+ = A^+$.

$$AA^+ A = (AA^+) A = \left(\sum_{k=1}^r v_k v_k^* \right) \left(\sum_{j=1}^r \sigma_j v_j v_j^* \right) = \sum_{j=1}^r \sigma_j v_j v_j^* = A \quad (33)$$

$$A^+ AA^+ = A^+ (AA^+) = \left(\sum_{k=1}^r \frac{1}{\sigma_k} u_k v_k^* \right) \left(\sum_{j=1}^r \sigma_j v_j v_j^* \right) = \sum_{k=1}^r \frac{1}{\sigma_k} u_k v_k^* = A^+ \quad (34)$$

Proof of Lemma 5.67

- when A has linearly independent columns, the rank is $r = n$. The set $\{u_1, \dots, u_n\}$ forms an orthonormal basis for C^n , so we have $\sum_{k=1}^n u_k u_k^* = I$. Thus

$$(A^* A)^{-1} = \left(\sum_{k=1}^n \sigma_k^2 u_k u_k^* \right)^{-1} = \sum_{k=1}^n \frac{1}{\sigma_k^2} u_k u_k^* \quad (35)$$

Multiplying by A^* , we obtain

$$(A^* A)^{-1} A^* = \left(\sum_{j=1}^n \frac{1}{\sigma_j^2} u_j u_j^* \right) \left(\sum_{k=1}^n \sigma_k u_k v_k^* \right) = \sum_{k=1}^n \frac{1}{\sigma_k} u_k v_k^* = A^+ \quad (36)$$

- when A has linearly independent rows, the rank is $r = m$. The set $\{v_1, \dots, v_m\}$ forms an orthonormal basis for C^m , so we have $\sum_{k=1}^m v_k v_k^* = I$. Thus

$$(AA^*)^{-1} = \left(\sum_{k=1}^m \sigma_k^2 v_k v_k^* \right)^{-1} = \sum_{k=1}^m \frac{1}{\sigma_k^2} v_k v_k^* \quad (37)$$

Multiplying A^* by this inverse, we obtain

$$A^* (AA^*)^{-1} = \left(\sum_{k=1}^m \sigma_k u_k v_k^* \right) \left(\sum_{j=1}^m \frac{1}{\sigma_j^2} v_j v_j^* \right) = \sum_{k=1}^m \frac{1}{\sigma_k} u_k v_k^* = A^+ \quad (38)$$